Exhaled Breath Sample Collection Protocol and Standard Operating Procedure (SOP)

ver. E1.0 (March 8, 2026)

1 Purpose

Standardize the collection process for exhaled breath samples from tuberculosis patients to ensure sample representativeness, integrity, and data reliability. By implementing standardized procedures, minimize environmental background interference and sampling errors to provide high-quality samples for subsequent analysis of exhaled breath volatile organic compounds (VOCs), biomarker screening, and disease diagnostics. Additionally, given the respiratory transmissibility of tuberculosis, this protocol aims to reduce biosafety risks during sampling, thereby protecting subjects, personnel, and the environment.

2 Scope of Application

This protocol applies to the collection of human exhaled breath samples from confirmed tuberculosis patients, suspected cases, or individuals with latent infections using FEP bags. It covers on-site sampling in various settings, including hospital respiratory outpatient clinics, tuberculosis-designated medical institution wards, community screening sites, and mobile sampling vehicles. The protocol encompasses both subject exhaled breath samples and corresponding environmental background samples.

3 Responsibilities

3.1 Sampling Personnel

(1) Prepare, operate, and maintain the FEP sampling bag equipment.

(2) Guide subjects through the exhalation procedures to ensure the quality of the samples.

3.2 Quality Control Personnel

(1) Supervise the collection process to ensure compliance with Standard Operating Procedure (SOP) requirements.

(2) Review the collected data, promptly correct any anomalies, and document the findings for archiving.

(3) Perform equipment calibration and monitor environmental parameters.

4 Work Procedure

4.1 Equipment and Materials

(1) Gas Bags: FEP sampling bags made of polyfluoroethylene propylene material; externally procured and tested prior to purchase.

(2) Protective Gear: Disposable gloves, disposable masks (N95 or medical surgical masks), medical protective gowns, disposable caps, shoe covers, and protective face shields or goggles (used during high-risk procedures).

(3) Disinfection Supplies: 75% alcohol, UV disinfection lamps, medical waste bags, and dedicated biosafety waste bins.

(4) Equipment: Breath gas analyzer (Proton Transfer Reaction Time-of-Flight Mass Spectrometer, model Jinxiezhuzun 20232070288).

(5) Auxiliary Tools:

a. Disposable mouthpieces (used if necessary to prevent cross-contamination).

b. Calibration gas (high-purity nitrogen).

c. Mixed calibration gases (10-component VOCs and 13-component VOCs for equipment calibration; purchased products with metrological certification).

d. Gas Mass Flow Controller.

e. Mechanical thermo-hygrometer

f. 50 mL glass syringe (for collecting environmental samples).

4.2 Sampling Procedure

4.2.1 Preliminary Preparation

4.2.1.1 Environmental Requirements

a. The sampling environment must be well-ventilated and situated upwind to prevent contamination from volatile organic compounds (VOCs). It should be located away from potential pollution sources such as air conditioner exhaust vents, kitchen exhaust ducts, parking garage exits, newly renovated areas, or chemical plants.

b. Biosafety Requirements: Sampling must be conducted in a dedicated, well-ventilated room.

4.2.1.2 Equipment and Material Inspection

a. Before testing, verify the proper operation of the exhaled breath analyzer. Calibrate the instrument regularly (e.g., weekly or monthly) using mixed standard gases, and document all calibration data thoroughly.

b. Check the gas bag's airtightness: Fill the gas bag with clean air or nitrogen to 80% of its rated capacity. Close the valve, gently press the gas bag to check for leaks, and leave it undisturbed for 30 minutes. If no significant volume reduction occurs, the gas bag is considered acceptable.

c. Verify that disposable mouthpieces (if required) are intact and individually packaged under sterile conditions. Ensure that the one-way valve at the gas bag inlet functions correctly.

4.2.1.3 Subject Preparation

(1) Day Before Sampling

a. Do not eat after 9:00 PM on the day before sampling (fasting).

b. On the evening before sampling, avoid alcohol (including alcoholic beverages), tea, coffee (including caffeinated drinks), energy drinks, and similar substances. Also, avoid foods with strong odors, such as garlic, onions, mint, bananas, tangerines, oranges, grapefruit, lemons, durian, and stinky tofu.

c. Avoid strenuous exercise and staying up late the day before sampling; maintain a regular sleep schedule.

(2) On the Day of Sampling

a. Rest quietly for 5 minutes before sampling and rinse your mouth with purified water (hold for 30 seconds), maintaining a normal breathing rate. Do not eat or drink anything after rinsing.

b. Smoking (including e-cigarettes) is prohibited on the day of sampling. Avoid using mouthwash, chewing gum, mint candies, or other volatile oral care products.

c. Health Status Disclosure: If the subject has respiratory conditions (e.g., asthma, acute pneumonia), metabolic disorders (e.g., diabetic ketoacidosis), or other health issues that may affect exhaled gas composition on the day of sampling, inform the sampling personnel in advance to facilitate accurate interpretation of results or to reschedule sampling.

4.2.1.4 Sampling Personnel Requirements

(1) Personal Protective Equipment (PPE) Protocol

In the clean area before the operation, put on the following items sequentially: a KN95 mask (perform a fit test), followed by a face shield or goggles, then a medical protective gown, and finally disposable gloves (tuck the glove cuffs over the gown sleeves).

(2) Verification

Connect the airbag to the disposable mouthpiece (if required), ensuring a secure seal. Before sampling, confirm that the airbag is fully deflated.

4.2.2 Collection Procedure

4.2.2.1 Subject Gas Sample Collection

a. Record the subject ID, name, sampling date, and time in the registration form or electronic system.

b. Exhalation Collection: The subject places the mouthpiece in their mouth (or directs breath directly into the bag inlet). After taking a deep breath, they exhale slowly, steadily, and continuously into the bag without pausing to breathe. Note: The sampler must ensure consistent exhalation force and speed (e.g., instructing the subject to “exhale steadily into the bag as if breathing normally”).

Avoid excessive force or pauses during exhalation.

c. Once the bag is full (reaching 80%–90% of its nominal capacity) or the subject is exhausted, immediately close the bag valve to prevent gas leakage or backflow.

d. Collected samples must be stored in light-protected conditions at room temperature (20–25°C) and tested within 72 hours.

4.2.2.2 Environmental Sample Collection

a. Sampling Point Selection: Select a location within the same environment as the subject, ideally at the environmental center point or within the subject's breathing zone. Avoid areas with potential interference, such as exhaust vents from air conditioner outdoor units, cafeteria fume ducts, parking garage exits, and zones with electromagnetic radiation.

b. Use a clean glass syringe, flushed with high-purity nitrogen, to draw environmental air from the selected central point.

c. Connect the syringe to a new, clean FEP gas bag labeled

d. Gently transfer the environmental air from the glass syringe into the gas bag, filling it to 80% capacity to avoid overfilling and prevent rupture.

e. After sampling, close the gas bag valve before removing the glass syringe to prevent gas backflow or leakage.

4.3 Operational Requirements

4.3.1 Biosafety and Personal Protection Requirements

4.3.1.1 Personal Protective Equipment (PPE) Requirements

a. All personnel entering the sampling area (including samplers, quality control personnel, and support staff) must strictly adhere to biosafety protection requirements.

b. Correctly wear N95 or KN95-grade respirators throughout the procedure and perform a fit test.

c. Wear disposable medical protective gowns, disposable caps, disposable gloves (double-layering is recommended), and shoe covers. When performing aerosol-generating activities (e.g., connecting tubing, handling sample bags), add a face shield or goggles.

d. Immediately replace any damaged, contaminated, or damp protective gear in a safe area.

e. After sampling, remove protective gear sequentially in the designated buffer zone, performing hand disinfection after each step. Never leave the contaminated area while wearing protective clothing.

4.3.1.2 Pre-Sampling Environmental Sterilization and Preparation

a. After daily testing concludes, irradiate the sampling room with ultraviolet lamps for at least 30 minutes, or wipe high-contact surfaces (e.g., desks, chairs, door handles) with 75% alcohol.

b. Ensure proper operation of the sampling area's ventilation system (e.g., negative pressure or high-efficiency filtration) to reduce airborne pathogen concentrations.

c. Prepare dedicated medical waste bins (yellow) and sharps containers, fitted with

double-layered yellow medical waste bags.

4.3.1.3 Operational Requirements During Sampling

a. All sample line connections must be performed gently to minimize aerosol generation.

b. Avoid direct bare-hand contact with gas bags or blowers whenever possible.

4.3.2 Equipment and Environmental Requirements

(1) Inspect the sample inlet tubing for cleanliness. Purge the tubing with high-purity nitrogen gas as necessary to ensure no residual VOCs remain.

(2) Confirm that the gas bag outlet is securely connected to the exhaled breath analyzer inlet, ensuring no leaks at the connection point.

(3) Maintain the sampling and analysis environment temperature between 20-25°C (68-77°F) with relative humidity ≤90% (record temperature and humidity values).

4.3.3 Sample Introduction Requirements

4.3.3.1 Stable Sampling

a. Connect the sample gas bag to the instrument inlet. Slowly and evenly squeeze the gas bag (or use its inherent pressure) to deliver gas at a constant flow rate, controlled by the flow controller, into the breath analyzer.

b. Ensure uniform gas discharge from the bag throughout the sampling period to prevent local concentration fluctuations caused by uneven compression.

c. Wait until the instrument's ion current signal stabilizes for 20 seconds (or until the real-time spectrum in the software stabilizes), then record the data. Remove the gas bag, close its valve, and conclude the sampling test.

4.3.3.2 Real-Time Monitoring

a. Continuously monitor the ion current signal in the breathalyzer software to ensure signal stability.

b. If significant signal fluctuations occur, immediately inspect the gas bag connection for leaks, adjust the squeezing pressure, or verify the gas flow rate settings.

4.3.4 Post-Test Environmental Sterilization and Cleaning

4.3.4.1 Sample and Waste Disposal

a. All disposable items that came into contact with the subject (e.g., mouthpieces, masks, gloves, tissues) and potentially contaminated items (e.g., outer packaging of gas bags) must be treated as medical waste. Place them in double-layered yellow medical waste bags, seal with a gooseneck knot, affix a "Biohazard" label, and dispose of them uniformly according to hospital infection control regulations.

b. Gas bag disposal: After testing, disinfect the outer surface of sample gas bags with ultraviolet irradiation, then dispose of the bags as medical waste.

4.3.4.2 Environmental and Equipment Disinfection

a. Surface disinfection: Immediately after sampling, wipe down all potentially contaminated surfaces—including sampling tables, instrument surfaces, chairs, door handles, etc.—with 75% alcohol. After 30 minutes, wipe clean with water to prevent equipment corrosion.

b. Air disinfection: After personnel exit, activate ultraviolet lamps to irradiate and disinfect the sampling room for at least 60 minutes. Alternatively, operate a mobile air disinfection unit for 1–2 hours.

c. Floor disinfection: Mop floors using a chlorine-based disinfectant.

d. Internal instrument tubing disinfection: For internal gas pathways of exhaled breath analyzers, consult the equipment manufacturer to determine appropriate methods (e.g., heated purging, UV irradiation) for periodic disinfection to prevent microbial growth within tubing.

4.4 Quality Control

(1) Blank controls: Use environmental air samples collected from the same batch as controls to analyze background VOC concentrations.

(2) Avoid cross-contamination: Each sample (including patient and environmental samples) must use a separate gas bag and disposable mouthpiece (if required). Reuse is strictly prohibited.

(3) System blank: After daily startup, analyze a high-purity nitrogen gas bag to verify the instrument system's cleanliness.

4.5 Sample Identification

Bag labels must include the bag number, sample type (patient or environmental), collection date (YYYY-MM-DD), and collection time (HH:MM). Use alcohol-resistant markers or waterproof printed labels, and ensure they are affixed with biohazard symbols.

4.6 Precautions

During transportation and temporary storage, gas bags must be protected from light exposure and crushing. They should be placed in dedicated containers or racks to prevent puncture by sharp objects and avoid high-temperature environments such as direct sunlight or the inside of vehicles. The outer surfaces of containers transporting samples must be disinfected and labeled with biohazard symbols.

4.7 Emergency Procedures

4.7.1 Bag Leakage

If leakage is detected during collection or sampling, immediately halt the process. If leakage occurs during collection, promptly assess the contaminated area, perform emergency disinfection of the affected zone, replace the bag with a new one, and restart the procedure. Document the incident.

4.7.2 Subject Discomfort

If a subject experiences dizziness, difficulty breathing, severe coughing, or other discomfort during exhalation, immediately pause the collection. Assist the subject in resting and provide necessary medical assistance. Then, determine whether to terminate the sampling or reschedule the collection.

4.7.3 Accidental Exposure

In cases of needle stick injuries or sample splashes onto skin or mucous membranes, immediately activate the occupational exposure emergency protocol: squeeze out blood from the wound, rinse with running water or saline, disinfect with povidone-iodine or alcohol, and report to supervisors and the Infection Control Department for medical observation and prophylactic treatment (if necessary).

5 Revision History

Version	Description of Changes	Prepared By	Date

6 Approval Records

Prepared		Date	
Reviewed		Date	
Approved		Date	